Sidney Lyayuga Lisanza

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Computational biochemist developing deep learning methods for protein design and AI-driven drug discovery. PhD-trained in biochemistry with experience spanning generative models, structure-conditioned design, and wet lab work.

EXPERIENCE

Machine Learning Scientist | Training Generative and Reasoning Models for Protein Design | **Prescient Design (Genentech)** | New York, NY | 2024-Present

- Develop generative and reasoning models for protein design that accelerate therapeutic discovery across small molecules, peptides, and large molecules.
- Partner with experimental collaborators to translate multi-objective therapeutic constraints into model requirements, followed by iterative design-build-test cycles with machine learning and wet lab teams to meet experimental endpoints.

Research Assistant | **Baker Lab, Institute for Protein Design, University of Washington** | Seattle, WA

- Developed generative machine learning methods for de novo protein design, with a focus on jointly modeling protein sequence and structure.
- Led work on RoseTTAFold sequence-space diffusion methods for controllable protein generation, culminating in ProteinGenerator and subsequent publication in Nature Biotechnology.
- Co-led development of early deep learning approaches for scaffolding functional sites in proteins, published in Science.

Undergraduate Researcher | **Baker Lab (UW) and Hahn Lab (UNC)** | Seattle, WA / Chapel Hill, NC

- Baker: Worked on computational protein design, including co-assembling multi-component protein crystals and early structure-guided design projects.
- Hahn: Studied GEF-H1 localization and activation dynamics using FRET-based biosensors and photoactivatable GEF inhibitors.

EDUCATION

PhD in Biochemistry | **University of Washington** | Seattle, WA

- Advisor: David Baker
- Thesis: Conditional Generation of Protein Sequence and Structure.

BA with Honors in Chemistry and Mathematics | **University of North Carolina at Chapel Hill** | Chapel Hill, NC

- Minor in Computer Science.

North Carolina School of Science and Mathematics

- Student Ambassador, Student Government, National Society of Black Engineers, NC Science and Engineering Fair, NC Student Academy of Science

SELECTED PUBLICATIONS

- Lab-in-the-loop therapeutic antibody design with deep learning. bioRxiv, 2025.
- Multistate and functional protein design using RoseTTAFold sequence space diffusion. Nature Biotechnology, 2025.
- Protein design using structure-prediction networks: AlphaFold and RoseTTAFold as protein structure foundation models. Cold Spring Harbor Perspectives in Biology, 2024.
- Joint generation of protein sequence and structure with RoseTTAFold sequence space diffusion. bioRxiv, 2023.
- Scaffolding protein functional sites using deep learning. Science, 2022.
- Deep learning methods for designing proteins scaffolding functional sites. bioRxiv, 2021.
- Design of proteins presenting discontinuous functional sites using deep learning. bioRxiv, 2020.
- Spatiotemporal dynamics of GEF-H1 activation controlled by microtubule- and Src-mediated pathways. Journal of Cell Biology, 2019.

TEACHING AND LEADERSHIP

- Co-organizer, GEM workshop at ICLR (2024, 2025, 2026): Workshop to foster collaboration between machine learning experts and experimental scientists.
- Teaching Assistant, UW BIOC 426 and UW BIOC 441: taught introductory biochemistry and core experimental techniques.
- Teaching Assistant, Rosetta REU Code School: taught undergraduate researchers how to design proteins with Rosetta.

TECHNICAL SKILLS

- Machine learning and programming: Python, PyTorch, Lightning, Prime-rl, Bash, C++, Java.
- Computational biology: protein design, generative modeling, sequence-structure modeling, scientific computing.
- Experimental biology: cloning, protein purification, protein characterization, and biochemical assay support.